

The Euclidean max–min problem for norm-one functionals has a sharp answer

Literature-status note for arXiv:1508.03518

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Status: literature_already_answered (one explicit subproblem). For arbitrary Euclidean norm-one functionals $f_1, \dots, f_m$ on $\mathbb{R}^n$, $n \geq 2$, one has the sharp universal estimate

$$\max_{\|x\|_2=1} \min_{1 \leq i \leq m} |f_i(x)| \geq \sin\left(\frac{\pi}{2m}\right).$$

The constant cannot be increased: equality holds when the normals are coplanar and equally spaced as unoriented lines. This follows immediately from Jiang and Polyanskii's 2017 proof of Fejes Tóth's zone conjecture.

The source question

Kobos asks for estimates of

$$M(f_1, \dots, f_m; \|\cdot\|) := \max_{\|x\|=1} \min_{1 \leq i \leq m} |f_i(x)|$$

for norm-one linear functionals on $\mathbb{R}^n$. The paper explicitly singles out the Euclidean norm as a first case and, by a measure argument, obtains the lower bound $1/(\sqrt{n}m)$ there [1, Section 3 and Problem 4]. The general-norm and projection-constant questions in the paper are separate and are not claimed to be settled here.

The later theorem

A zone of width ω on the unit sphere is the set of points within spherical distance $\omega/2$ of a great sphere. Jiang and Polyanskii prove that any finite collection of zones covering a unit sphere has total width at least π ; their theorem holds in every sphere dimension and also characterizes equality [2, Theorem 1].

Represent f_i by a unit normal u_i , so $f_i(x) = \langle u_i, x \rangle$. Put

$$q := \max_{x \in S^{n-1}} \min_i |\langle u_i, x \rangle|.$$

For each i , define

$$Z_i := \{x \in S^{n-1} : |\langle u_i, x \rangle| \leq q\}.$$

The spherical distance of x from the great sphere $u_i^\perp \cap S^{n-1}$ is $\arcsin |\langle u_i, x \rangle|$. Consequently Z_i is a zone of width $2 \arcsin q$. The definition of q says exactly that the m zones Z_i cover S^{n-1} . The zone theorem therefore gives

$$2m \arcsin q \geq \pi, \quad \text{hence} \quad q \geq \sin\left(\frac{\pi}{2m}\right).$$

Sharpness

Fix a two-dimensional subspace $L \subset \mathbb{R}^n$, and choose unit normals $u_1, \dots, u_m \in L$ whose unoriented directions are equally spaced around the projective circle, with successive angle π/m . For any unit x , write $r = \|\text{proj}_L x\|_2$. If $r = 0$, all the functionals vanish. Otherwise, some great-line direction $u_i^\perp \cap L$ lies within angle $\pi/(2m)$ of $\text{proj}_L x/r$. For that i ,

$$|f_i(x)| = r |\langle u_i, \text{proj}_L x/r \rangle| \leq r \sin\left(\frac{\pi}{2m}\right) \leq \sin\left(\frac{\pi}{2m}\right).$$

Thus q is at most the asserted constant for this configuration, while the preceding lower bound applies to every configuration. Equality follows.

Equivalently, if one minimizes $M(f_1, \dots, f_m; \|\cdot\|_2)$ over all Euclidean norm-one m -tuples, its exact value is $\sin(\pi/(2m))$, independently of $n \geq 2$. Compared with the source estimate, this is asymptotic to $\pi/(2m)$ and removes the dimension loss.

Scope and verification

The source problem and its Euclidean estimate were checked in the official arXiv source and PDF. The covering theorem and its all-dimensional scope were checked in the official PDF of arXiv:1703.10550v2, especially Theorem 1. The argument above is a direct translation between absolute functional values and angular distances from great spheres; no additional regularity or general-position hypothesis is used.

References

- [1] Tomasz Kobos, *A uniform estimate of the relative projection constant*, arXiv:1508.03518; J. Approx. Theory **225** (2018), 58–75.
- [2] Zilin Jiang and Alexandr Polyanskii, *Proof of László Fejes Tóth’s zone conjecture*, arXiv:1703.10550v2; Geom. Funct. Anal. **27** (2017), 1367–1377. DOI: 10.1007/s00039-017-0427-9.